Andreas Stigsbøl Krühlmann

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Profile

I am a software engineer with experience in full-stack development, system architecture, Linux infrastructure and deployment automation. I take responsibility across the software lifecycle, from gathering requirements and making architectural decisions to implementation, deployment and ongoing maintenance. I work closely with stakeholders to translate business needs into practical technical solutions, with a focus on maintainable code, automated testing and observability. Alongside hands-on development, I support colleagues through mentoring, code reviews and collaborative work through pair-programming.

Technical skills

Languages & Frameworks: Rust, Python, TypeScript, JavaScript, Java, SQL, FastAPI, Flask, React, Svelte, WebAssembly, Spring Boot

Infrastructure: Linux, Docker, Docker Swarm, Ansible, PostgreSQL, Redis, MQTT, InfluxDB, Grafana, Virtualization

AI: Azure OpenAI, vector search, llama.cpp, strata, self-hosting

Methods

Development: Domain-Driven Design, Behaviour-Driven Development, Test-Driven Development

Delivery & Planning: CI/CD, DevOps, Scrum, Kanban, Infrastructure as Code

Collaboration Mentoring, pair-programming, open-source development

Experience

Software Engineer 2023–2026
Siemens Mobility A/S

- Developed the Copenhagen Light Rail departure-screen system, initially as sole developer. Took responsibility for requirements, architecture, backend services, display software, Linux clients, deployment and support. The production system served more than 100 client devices across all 29 stations.
- Built the solution using Rust, WebAssembly and containerised backend services, with a React/TypeScript management interface. Implemented remote configuration, automatic brightness adjustment, monitoring and diagnostics.
- Introduced centrally managed Linux clients booting over PXE from a read-only NFS filesystem, enabling updates without station visits and recovery to a known software state on reboot.
- Led development of a fleet-wide telemetry platform collecting onboard train data through MQTT. Implemented ingestion and storage in InfluxDB, Grafana dashboards and automated threshold alerts to support maintenance staff.
- Co-designed and delivered practical AI training for approximately 50 technical and nontechnical colleagues. Developed an Azure OpenAI prototype using embeddings and vector search to retrieve internal process documentation, working directly with nontechnical stakeholders.
- Onboarded and mentored two student developers, supporting delivery through code reviews and technical guidance.

Experience continued

Data Technician Apprentice 2019–2023
Siemens Mobility A/S

- Co-developed a customer-facing railway KPI dashboard covering punctuality, technical faults, maintenance and contractual reporting. Implemented data ingestion, calculations, APIs and frontend functionality using Python, FastAPI, PostgreSQL and Svelte.
- Used database indexing, partitioning and precomputed aggregates to support responsive reporting across large datasets and a 30-year retention requirement. The application became Siemens Mobility Denmark's first internally developed software service delivered to a customer.
- Designed and implemented Ansible automation for on-premises infrastructure supporting Oslo metro applications. Automated server, storage, networking, firewall and Docker configuration, enabling internal deployments without third-party participation.
- Supervised two student developers delivering a vehicle booking and mileage-tracking application. Gathered requirements, made architecture decisions, reviewed code and coordinated delivery. The application replaced paper vehicle logs with shared digital records.

Student Worker 2015–2019
Siemens Mobility A/S

- Co-developed an employee onboarding application adopted across all Siemens companies in Denmark. Contributed database design, frontend development, authentication, personal-data anonymisation, deployment and maintenance.
- Co-developed and maintained a training and compliance management application using Python, Flask and PostgreSQL. Implemented APIs, business logic, reporting and evidence exports, replacing spreadsheets and manual records.
- Maintained and extended Excel/VBA automation used daily by approximately ten financial controllers, adding data imports and interactive workbook features.

Additional experience

Technical Author July 2015–March 2017
Packt Publishing

Co-authored *Building a 3D Game with LibGDX*. Wrote technical chapters and developed the accompanying Java example game and source code. Contributed deployment instructions, technical review and editing.

Freelance Full-Stack Developer March 2012–August 2013
Copenhagen Judo

Sole developer of a PHP/MySQL website supporting tournament registration and attendance tracking for the Danish national judo team.

Education

Data Technician, specialisation in programming 2023
TEC, Ballerup

Languages

Danish: Native English: Fluent